

Task 1: Access the GRUB2 Boot Menu

Steps:

1. Reboot the VM.
 - `sudo reboot`
2. During startup:
 - BIOS VM: Press and hold Shift.
 - UEFI VM: Press Esc repeatedly.
3. The GRUB2 Boot Menu appears.

Task 2: Find when the system reached multi-user.target

View current boot logs

- `journalctl -b`

Search for "Reached target"

- `journalctl -b | grep "Reached target"`

Search specifically for Multi-User Target

- `journalctl -b | grep "multi-user.target"`

Task 3: Simulate a Boot Failure Using /etc/fstab

Steps:

1. Backup fstab:
 - `sudo cp /etc/fstab /etc/fstab.bak`
2. Edit the file:
 - `sudo vi /etc/fstab`
3. Save and reboot:
 - `sudo reboot`

Recover

At the emergency shell:

- `mount -o remount,rw /`

Restore the backup:

- `cp /etc/fstab.bak /etc/fstab`

Reboot:

- `reboot`

Task 4: Identify Failed Services

List failed services:

- `systemctl --failed`

Check the status of a failed service:

- `systemctl status <service-name>`

View detailed logs:

- `journalctl -xeu <service-name>`

Task 5: Recovery Plan for Emergency Shell

Steps:

1. Log in as root.
2. Mount the root filesystem as read/write.
 - `mount -o remount,rw /`
3. Check `/etc/fstab`.
 - `cat /etc/fstab`
4. Save the file.
5. Reload systemd configuration.
 - `systemctl daemon-reload`
6. Reboot the system.
 - `reboot`